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5 Questions with a Medical AI Expert



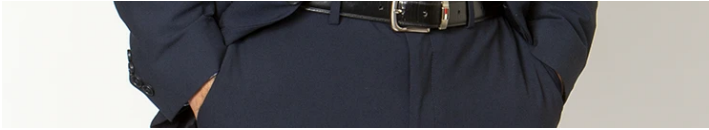
A conversation with Pranav
Rajpurkar

Sleep Issue

 Ekaterina Pesheva

—  3 minute read

—  Interview



Pranav Rajpurkar
Assistant professor
of biomedical
informatics in the
Blavatnik Institute at
HMS

What sparked your interest in artificial intelligence and in medical AI specifically?

During freshman orientation at Stanford, I heard Andrew Ng from Stanford's artificial intelligence lab give a talk on AI. He spoke with such enthusiasm about the effects AI would have on our lives. I emailed him, telling him of my hope to get involved in AI research. He invited me to join his lab, which I did during my freshman year. I stayed for the rest of my undergrad career as part of a team building artificial intelligence for autonomous driving. His lab was a playground with a lot of talented people I worked with and learned from. I found my calling during the second year of my PhD program when I worked at the intersection of AI and medicine: I started building artificial intelligence tools to detect abnormal heart rhythms in electrocardiograms and

later built AI tools across different medical specialties for disease diagnosis, risk prediction, and therapy recommendations.

What do you see as the greatest promises and hurdles of medical AI?

The central promise of medical AI — which is also the mission of my lab — is the idea that AI can safely automate many clinical decision-making tasks. This should help improve patients' lives. The quest to develop high-performance medical AI algorithms that can help doctors in their work is an important pursuit. But designing the consummate AI doctor will require us to tackle challenges on three fronts: algorithm design, dataset curation, and implementation design. Right now, high-performance algorithms are built using human-labeled data, but in the future, we will need to leverage the vast swaths of unlabeled data when designing these algorithms. On the dataset front, we must build ones that work reliably for a variety of patient populations and clinical settings worldwide. For implementation design, the focus rests on enhancing the interaction between the AI algorithm and the clinician so that the AI tools allow clinicians to make optimal decisions. We need to design these systems to extract the good while safeguarding against the bad.

What is on your to-do list?

One of my main interests is creating communities around AI and medicine. At HMS, I founded the Medical AI Bootcamp, a joint Harvard–Stanford mentored research program for college students that involves AI in medicine. I also try to bring people into medical AI by curating materials that appeal to any level of interest and background. I’ve taught an online course series on AI and medicine, I co-host the *AI Health Podcast*, and I co-write *Doctor Penguin*, an AI and health care research newsletter.

Who are your heroes?

My parents instilled in me a love of science and reading at an early age. My love of research and teaching was shaped by several advisors in AI and computer science, including Ng, Percy Liang, and Michael Bernstein at Stanford, and my longtime mentors in medical AI at Stanford, Matt Lungren, Curt Langlotz, and Eric Topol, the Scripps Research Translational Institute founder.

What do you do outside of work?

I enjoy walking around Boston, mostly to coffee shops. I also like to cook and try new recipes. I’m currently in the groove of weight lifting but occasionally enjoy golf and tennis.

Ekaterina Pesheva is senior director of science communications and media relations in the HMS Office of Communications and External Relations.

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